

Ziqiang Xu

Post-Master Researcher

Ulsan National Institute of Science and Technology (UNIST)

xuziqiang@unist.ac.kr

 github  website

Research Interests

- Artificial Intelligence
- Remote Sensing
- Deep Learning for Computer Vision & Object Detection and Recognition
- NLP & Computer Security (Kali Linux) **Extracurricular Interests*

Education

Kyungpook National University <i>Graduate School, M.S.</i> <i>Robot and Smart System Engineering & AI-Converged Safety</i> <i>GPA: 4.02/4.3</i> <i>Supervisor: Bubryur Kim</i>	Mar. 2024 – Mar. 2026 <i>Daegu, South Korea</i>
North University of China <i>College of Information and Business, B.S.</i> <i>Computer Science and Engineering</i> <i>GPA: 3.35/4.0</i> <i>Thesis Supervisor: Yingliang Zhao</i>	Sep. 2019 – Jun. 2023 <i>Jinzhong, China</i>

Academic Experience

Post-Master Researcher <i>Ulsan National Institute of Science and Technology (UNIST)</i> <i>Department of Civil, Urban, Earth, and Environmental Engineering</i> Ulsan, South Korea	Mar. 2026 – Present
Research Intern <i>Ulsan National Institute of Science and Technology (UNIST)</i> <i>Department of Civil, Urban, Earth, and Environmental Engineering</i> Ulsan, South Korea	Dec. 2025 – Feb. 2026
Research Assistant Grade 3 <i>Kyungpook National University, Daegu, South Korea</i>	Sep. 2025 – Feb. 2026
BK21 Research Assistant <i>Kyungpook National University, Daegu, South Korea</i>	Mar. 2025 – Aug. 2025
Research Intern <i>Intelligent Construction Automation Center, Daegu, South Korea</i>	Nov. 2023 – Mar. 2024
Undergraduate Intern <i>Artificial Intelligence Lab, Jinzhong, China</i>	Jun. 2021 – Jun. 2023

Publication

1. Satellite-Based Wildfire Monitoring Across the Fire Lifecycle Using Artificial Intelligence: Trends, Challenges, and Future Directions
Accepted (2026)
2. GreenViT: A Vision Transformer with Single-Path Progressive Upsampling for Urban Green-Space Segmentation and Auditable Area Estimation
Accepted (2026)
3. Trustworthy Celestial Eye: Calibrated and Robust Planetary Classification via Self-Supervised Vision Transformers
Accepted (2026)
4. Deep Convolutional Stacked Model for Material Industry Applications: Automated Precise Porosity Assessment Based on CT Imaging
Accepted for presentation (2025)
5. Concrete Structure Crack Classification and Analysis Using Neural Network and Clustering
Accepted (2024)
6. Application of Deep Learning Model Based on ResNet-50 Architecture in Semantic Segmentation of Satellite Images
Accepted for presentation (2024)

Honors & Awards

- 2024–2026 KINGS Scholarship, Kyungpook National University
- Sep. 2025 Research Assistantship Grade 3, Kyungpook National University
- Mar. 2025 BK21 Research Assistantship, Kyungpook National University
- Sep. 2024 Excellence Award, Startup Idea Competition in Integrated Safety Engineering, Graduate School of Convergence Technology in Land and Transport Safety
- Jan. 2024 Merit-based Scholarship, School of Korean Language Education, Kyungpook National University
- 2021-2022 Third Prize, Lanqiao Cup National Software and IT Talent Competition(Python Group), Ministry of Industry and Information Technology of China

Technical Skills

Programming Languages: Python, HTML, CSS

Developer Tools: PyCharm, Visual Studio Code, Cursor

Frameworks / Libraries: PyTorch, Numpy(Pandas), OpenCV(cv2), TensorFlow, Git, LaTeX, Kali Linux

Language Proficiency

- **Chinese:** Native
- **Korean:** Fluent (TOPIK Level 5)
- **English:** Basic proficiency (CET-6)